

Question

In the ion $[\text{Fe}_5\text{O}(\text{CH}_3\text{COO})_{12}]^+$, iron exhibits two coordination numbers and two chemical environments, with a ratio of 2 : 3. The acetate groups act as bridging or monodentate ligands.

Among the following statements about the structure of this compound, identify the correct one.

- A.** In this complex, the O monoatomic ligand is connected to four Fe atoms.
- B.** In this complex, there is a 5-coordinated Fe.
- C.** In this complex, a five-membered ring structure is present.
- D.** In this complex, the bridging acetate ligand has only one chemical environment.
- E.** There are no atoms at the geometric center of this complex ion.
- F.** In this complex, there exists a Y-shaped structure formed with the participation of Fe atoms, which is coplanar or nearly coplanar.
- G.** In this complex, if all freely rotatable parts connected by single bonds are considered spherical and minor deviations are neglected, the order of the point group to which the complex belongs is 3.
- H.** In this complex, the monodentate acetate ligand and the monatomic O ligand can occupy adjacent positions in the coordination polyhedron of the Fe atom.
- I.** None of the above options is correct

Answer

Correct Answer: **F**

Detailed Explanation

Examining the composition of $[\text{Fe}_5\text{O}(\text{CH}_3\text{COO})_{12}]^+$, there is one single-atom O ligand, which is preferentially placed at the center. The five Fe atoms are divided into two chemical environments in a 2:3 ratio, likely forming a Fe_5O trigonal bipyramidal structure around O, where two Fe atoms occupy axial positions and three Fe atoms occupy equatorial positions.

CHECKPOINT

1 PTS

Fe_5O trigonal bipyramid exists

Among the 12 acetate ligands, there are two coordination modes: monodentate and bridging. Due to the core $\text{Fe} - \text{O} - \text{Fe}$ structure, the acetate ligands can form stable six-membered rings when bridging. Therefore, when arranging the acetate ligands, bridging should be prioritized.

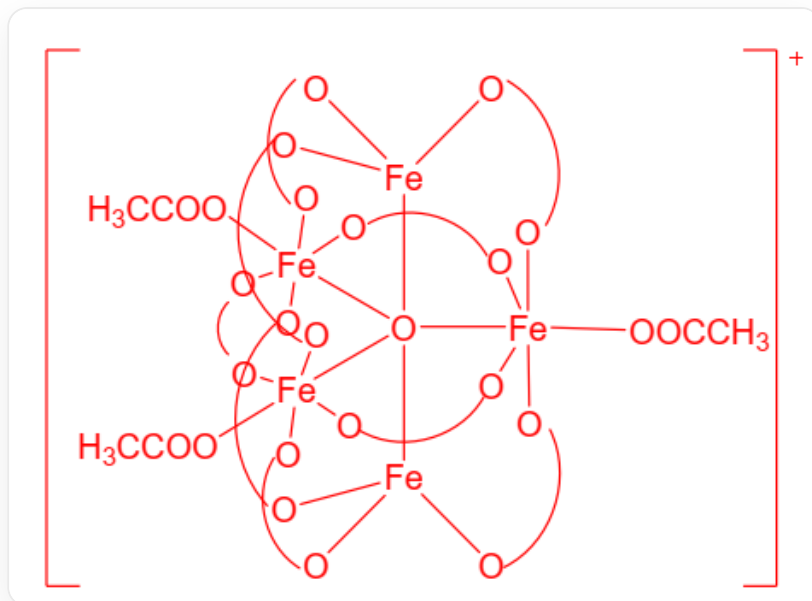
CHECKPOINT

1 PTS

Bridging acetates can participate in forming six-membered rings

Within the trigonal bipyramid, excluding the axial pair of Fe atoms, the five Fe atoms can form nine adjacent FeFe pairs: three in the equatorial plane and six along the meridional direction. If each FeFe pair is bridged by one acetate, nine acetates are consumed, leaving three terminal acetates, which can be placed on the equatorial Fe atoms.

The resulting structure of the $[\text{Fe}_5\text{O}(\text{CH}_3\text{COO})_{12}]^+$ ion is as follows:



O is at the center, with five Fe atoms forming a trigonal bipyramid around it. The two axial Fe atoms are tetrahedrally 4-coordinated, while the three equatorial Fe atoms are octahedrally 6-coordinated. On the coordination polyhedra of the equatorial Fe atoms, the position opposite to the central O is occupied by terminal OOCCH_3 groups—three in total. There are nine bridging $\text{O} \sim \text{O}$ groups, each connecting two different Fe atoms: three in the equatorial plane, linking the equatorial Fe atoms pairwise, and six along the meridional direction, connecting axial and equatorial Fe atoms.

In the figure, $\text{O} \sim \text{O}$ represents bridging acetate groups. It can be seen that the five Fe atoms form a trigonal bipyramid around the central O, with two chemical environments: three 6-coordinated Fe atoms and two 4-coordinated Fe atoms, along with three terminal acetates and nine bridging acetates, satisfying the given conditions.

CHECKPOINT

1 PTS

There are three 6-coordinated Fe atoms, two 4-coordinated Fe atoms, three terminal acetates, and nine bridging acetates

In this complex structure, a Fe_5O trigonal bipyramid exists, with the oxygen atom being 5-coordinated, so **A** is incorrect.

This complex has three 6-coordinated Fe atoms and two 4-coordinated Fe atoms, with no 5-coordinated Fe atoms, so **B** is incorrect.

In this complex, the smallest ring is $\text{O}_{\text{central}} - \text{Fe} - \text{O}_{\text{acetic}} - \text{C} - \text{O}_{\text{acetic}} - \text{Fe}$, a six-membered ring, so **C** is incorrect.

CHECKPOINT

1 PTS

The smallest ring is a six-membered ring

In this complex, the nine bridging acetates are divided into two types: three in the equatorial plane and six along the meridional direction, with different chemical environments, so **D** is incorrect.

CHECKPOINT

1 PTS

Bridging acetates are divided into three in the equatorial plane and six along the meridional direction

From the complex structure, it can be seen that the geometric center is the 5-coordinated O atom, so **E** is incorrect.

CHECKPOINT

1 PTS

The geometric center is the 5-coordinated O atom

Extracting the equatorial Fe_3O from the Fe_5O trigonal bipyramid reveals an approximately coplanar Y-shaped structure. Thus, **F** is correct.

CHECKPOINT

1 PTS

In the Fe_5O trigonal bipyramid, the equatorial direction forms an approximately coplanar Y-shaped Fe_3O

From the complex structure, the core framework is a Fe_5O trigonal bipyramid, and all acetate ligands are symmetrically distributed (ignoring minor rotational orientations and atomic deviations). Thus, the overall point group is D_{3h} with an order of 12, so **G** is incorrect.

CHECKPOINT

1 PTS

The point group is D_{3h}

In this complex structure, the monodentate acetates are located at the positions opposite to the single-atom O ligand on the octahedrally 6-coordinated Fe atoms, so **H** is incorrect.

CHECKPOINT

1 PTS

The monodentate acetates are located at the positions opposite to the single-atom O ligand on the octahedrally 6-coordinated Fe atoms

In conclusion, the correct option is **F**.